

Estimating Deaton’s unit-value model: the `duvm` Stata module

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Software: `duvm` 1.0.0, <https://github.com/aabdd12/duvm>

Abstract

`duvm` estimates the unit-value model of Deaton (1988, 1990, 1997): own- and cross-price elasticities of quantity, expenditure elasticities and quality elasticities for a group of goods, from the budget shares and unit values that households report, using the spatial variation of prices across clusters. The estimator is Deaton’s two-stage procedure, closed-form and written in Mata: within-cluster regressions, between-cluster regressions of the cluster averages with the errors-in-variables correction, the quality parameter derived from weak separability, completion of the system with a composite good, and an approximate Slutsky symmetry. `duvm` adds what the published programs lack: sampling weights with the effective cluster sizes they imply; a linearized variance built from the influence functions of every first- and second-stage moment, including the generated-regressor terms of the first stage, combined through the analytic Jacobian of the chain from the moments to the elasticities, and aggregated by cluster or by survey design (`svyset`: PSUs, strata, finite-population correction); a design bootstrap of both stages; a diagnostic of the price signal before estimation; standard `e()` results and `estat`. Every variance is validated against the bootstrap under simple random sampling of clusters, sampling weights and a stratified design: the linearized standard errors are within ten percent of the bootstrap for all 34 parameters. An option `compat` reproduces the formulas of the earlier WELCOM implementation, which followed the Stata code published with Deaton (1997); Appendix A records the four places where that code departs from the book and the effect of each.

Keywords: unit values, quality shading, price elasticities, household surveys, errors in variables, survey design, influence functions.

1 Introduction

Household surveys in developing countries rarely record prices, but they record expenditures and quantities, whose ratio, the unit value, carries the spatial variation of prices between villages. Deaton (1988) showed how to turn this into price elasticities of demand once two contaminations are removed: the unit value also reflects the quality chosen by the household, which rises with its income and falls when the price rises, and it is measured with an error correlated with the error in the quantity. His model, developed in Deaton (1990) and laid out in full in chapter 5 of Deaton (1997), remains the standard tool whenever price data are missing: a first stage within clusters, where prices are constant, identifies the income and household effects; a second stage between clusters, on the averages purged of those effects, identifies the price responses; weak separability of preferences turns the quality elasticity with respect to income into the quality response to price; and an errors-in-variables correction uses the first-stage residuals to purge the between-cluster moments.

The software has not followed the method. Deaton (1997) published the Stata programs of the Maharashtra and Pakistan applications — `allindia.do`, `mkmats.do`, `bootall.do` — as an appendix on the World Bank’s site; they are Stata 4 code tied to one data set, without weights, and their only standard errors come from a bootstrap of the second stage alone. The WELCOM package (Araar and Olivieri, 2018) wrapped that code into one command, `duvm`, and added sampling weights, a selection correction and results by group; its formulas are those of the published programs. We know of no other packaged implementation in Stata, R or Python; the applied literature re-implements the method from the paper each time, or falls back on unit values used as prices.

`duvm` 1.0 is a new implementation of the model as the book states it, in Mata, with the inference the method deserves. Section 2 states the model and the parameters that are identified. Section 3 gives the estimator, stage by stage, with the exact formulas used, including the treatment of weights. Section 4 derives the linearized variance and describes the bootstrap. Section 5 presents the diagnostic. Section 6 documents the command and reports the results on the four cereal groups of the Mexican ENIGH 2014, which ship with the package. Section 7 validates the variance. Appendix A documents the differences between the published code and the book, with their measured effect, and the option that reproduces the earlier results.

2 The model

Households h live in clusters c ; within a cluster all households face the same prices π_{Hc} of the M goods $H = 1, \dots, M$. Household h has total expenditure x_{hc} , characteristics z_{hc} , and for each good G a budget share w_{Ghc} and, when it buys the good, a unit value $v_{Ghc} = \text{expenditure} / \text{physical quantity}$. Following Deaton (1997, eqs. 5.76–5.77),

$$w_{Ghc} = \alpha_G^0 + \beta_G^0 \ln x_{hc} + \gamma_G^{0'} z_{hc} + \sum_{H=1}^M \theta_{GH} \ln \pi_{Hc} + f_{Gc} + u_{Ghc}^0, \quad (1)$$

$$\ln v_{Ghc} = \alpha_G^1 + \beta_G^1 \ln x_{hc} + \gamma_G^{1'} z_{hc} + \sum_{H=1}^M \psi_{GH} \ln \pi_{Hc} + u_{Ghc}^1. \quad (2)$$

The share equation has a cluster effect f_{Gc} uncorrelated with prices; the unit-value equation has none, since conditional on price a unit value reflects only quality and measurement error. β_G^1 is the expenditure elasticity of quality; ψ_{GH} is the response of the unit value to prices, unity on the diagonal in the absence of quality shading; θ_{GH} is the response of the budget share. The error terms include the measurement errors in quantities and unit values, and u^0 and u^1 are correlated within a household.

The quantities of interest follow from the identities $w = qv/x$ and $\ln v = \ln \pi + \ln \xi$, where ξ is quality. Writing $\bar{w}_G$ for the mean budget share,

$$e_G = 1 - \beta_G^1 + \beta_G^0 / \bar{w}_G \quad (3)$$

is the total expenditure elasticity of quantity, and the matrix of price elasticities of quantity is

$$E = -\Psi + D(\bar{w})^{-1}\Theta, \quad (4)$$

(Deaton, 1997, eqs. 5.88–5.89). Neither Θ nor Ψ is identified from the data alone: the between-cluster regression of shares on unit values estimates only $B = (\Psi')^{-1}\Theta'$, eq. (5.86). Weak separability of preferences within each group supplies the missing restriction: the quality of good G depends on the price of H only through the quantity of G , so that

$$\psi_{GH} = \delta_{GH} + \frac{\beta_G^1 e_{GH}}{e_G}, \quad \Psi = I + D(\beta^1) D(e)^{-1} E, \quad (5)$$

eqs. (5.75) and (5.87). Combining (4) and (5) with $\Theta = B'\Psi$ gives

$$\Psi = [I - D(\zeta)B' + D(\zeta)D(\bar{w})]^{-1}, \quad \Theta = B'\Psi, \quad E = [D(\bar{w})^{-1}B' - I]\Psi, \quad (6)$$

where the quality parameter of good G is

$$\zeta_G = \frac{\beta_G^1}{(1 - \beta_G^1)\bar{w}_G + \beta_G^0} = \frac{\beta_G^1}{\bar{w}_G e_G}, \quad (7)$$

eqs. (5.90)–(5.92). The product $\zeta_G \bar{w}_G = \beta_G^1 / e_G$ is the share of a price change absorbed by quality shading; when it is small, Ψ is close to the identity and the unit values behave as prices.

Completing the system. The M goods with unit values do not exhaust the budget. Adding-up and homogeneity extend Θ to an $(M+1) \times (M+1)$ matrix Θ^x whose last column is $-\sum_H \theta_{GH} - \beta_G^0$ and whose last row makes every column sum to zero, eqs. (5.93)–(5.94); $\bar{w}$, β^0 extend by adding-up, and the composite good receives an assumed quality elasticity β_{M+1}^1 (0.25 in Deaton's code; `qother()`). With ζ^x from (7) on the extended vectors,

$$\Psi^x = [I + D(\zeta^x)D(\bar{w}^x)]^{-1}[I + D(\zeta^x)\Theta^x], \quad B^{x'} = \Theta^x(\Psi^x)^{-1}, \quad E^x = [D(\bar{w}^x)^{-1}B^{x'} - I]\Psi^x, \quad (8)$$

eqs. (5.95)–(5.97). The $M \times M$ block of E^x equals E : completion adds the composite good without disturbing the others.

Symmetry. Slutsky symmetry requires $\theta_{GH} + \beta_G^0 \bar{w}_H = \theta_{HG} + \beta_H^0 \bar{w}_G$, eq. (5.71). Because it is nonlinear in B once quality enters, Deaton imposes it approximately, on $B + \bar{w}\beta^{0'}$, which coincides with the exact restriction when $\Psi = I$. With R the matrix that picks the $M(M-1)/2$ pairs of $\text{vec } B$ and $r = -R(\beta^0 \otimes \bar{w})$, the restricted estimate is

$$\text{vec } \tilde{B}_R = \text{vec } \tilde{B} + (I \otimes A)^{-1} R' [R(I \otimes A)^{-1} R']^{-1} (r - R \text{vec } \tilde{B}), \quad (9)$$

with A the corrected moment matrix of the second stage, eq. (5.102). `duvm` reports the elasticities with and without the restriction.

3 Estimation

3.1 First stage: within clusters

Each of the $2M$ equations is estimated by weighted least squares on the variables demeaned within cluster (the fixed-effect estimator), the unit-value equations on the households that report a unit value, the share equations on all households. With household weights ω_h , the cluster mean of a variable is $\sum_{h \in c} \omega_h y_h / \sum_{h \in c} \omega_h$ and the estimator is $\hat{\beta} = (\tilde{X}' \Omega \tilde{X})^{-1} \tilde{X}' \Omega \tilde{y}$. The first stage yields $\hat{\beta}_G^0$, $\hat{\beta}_G^1$, the coefficients of the characteristics, and the residuals e_{Ghc}^0 , e_{Ghc}^1 . From the residuals,

$$\hat{\omega}_G = \frac{\kappa_G}{n_G^+ - k - C_G^+} \sum_h \omega_h m_{Gh} (e_{Gh}^1)^2, \quad \hat{\chi}_G = \frac{\kappa_G}{n_G^+ - k - C_G^+} \sum_h \omega_h m_{Gh} e_{Gh}^0 e_{Gh}^1, \quad (10)$$

are the variance of u^1 and the covariance of u^1 and u^0 , eqs. (5.81)–(5.82), where m_{Gh} marks the reporters, n_G^+ and C_G^+ count the reporting households and clusters, k the regressors, and $\kappa_G = n_G^+ / \sum \omega_h m_{Gh}$ normalises the weights as Stata's analytic weights do. The purged series

$$\hat{y}_{Ghc}^0 = w_{Ghc} - x_{hc}' \hat{\beta}_G^0, \quad \hat{y}_{Ghc}^1 = \ln v_{Ghc} - x_{hc}' \hat{\beta}_G^1, \quad (11)$$

keep the cluster effects and the price terms, eqs. (5.78)–(5.79); x_{hc} collects log expenditure, log household size and the characteristics.

3.2 Second stage: between clusters

The cluster averages $\bar{y}_{Gc}^0$ (over all households) and $\bar{y}_{Gc}^1$ (over the reporters), weighted like the first stage, are residualised on the constant and the region and survey-round dummies when these are given (the *purge*, unweighted across clusters). Their between-cluster covariances,

$$S_{GH} = \text{Cov}_c(\bar{y}_{Gc}^1, \bar{y}_{Hc}^1), \quad R_{GH} = \text{Cov}_c(\bar{y}_{Gc}^1, \bar{y}_{Hc}^0), \quad (12)$$

are computed pairwise over the clusters where both series are observed. They contain the averaged first-stage errors, ω_G/n_{Gc}^+ and χ_G/n_{Gc} , which do not vanish when clusters are small. Deaton's errors-in-variables estimator removes them with the harmonic means of the cluster sizes, eq. (5.55):

$$\tilde{B} = (S - D(\hat{\omega}) \bar{N}_+^{-1})^{-1} (R - D(\hat{\chi}) \bar{N}^{-1}), \quad \bar{n}_G = \left(\frac{1}{C} \sum_c n_{Gc}^{-1} \right)^{-1}, \quad (13)$$

eq. (5.85). With weights, the variance of a weighted cluster mean of n independent errors is $\sigma^2 \sum_h \omega_h^2 / (\sum_h \omega_h)^2$, so the size of a cluster is its effective size $n_c^{\text{eff}} = (\sum_h \omega_h)^2 / \sum_h \omega_h^2$ (Kish, 1965), which equals the number of households when the weight is constant within the cluster — the usual case, and the case of the Mexican data. This is the single point at which the weights change the second stage; the second stage itself treats each cluster as one observation, since each cluster is one draw of the price vector.

From $\tilde{B}$, the symmetry restriction (9) with $A = S - D(\hat{\omega}) \bar{N}_+^{-1}$, then (7), (6) and (8) give the elasticities. Everything is closed-form; on 15,866 households, 2,622 clusters and four goods the whole chain takes a fifth of a second (Table 6).

3.3 Non-buyers

A household that does not buy a good has a zero share and no unit value. The share equation is estimated on all households, the unit-value equation on the buyers; the purged unit-value average of a cluster without buyers is missing and the cluster drops out of the pairs that involve that good. Option `csb(1)` adds to each share equation the inverse Mills ratio of a probit of buying the good on the first-stage regressors, the correction of the WELCOM version; the Mills ratio is treated as a fixed regressor in the variance.

4 Variance

4.1 Linearization

The estimator is a smooth function $\theta = g(\eta)$ of the vector of moments

$$\eta = (\text{vec } S, \text{vec } R, \hat{\omega}, \hat{\chi}, \hat{\beta}_{\cdot,1}^0, \hat{\beta}_{\cdot,1}^1, \bar{w}),$$

of dimension $2M^2 + 5M$, where $\hat{\beta}_{\cdot,1}^j$ are the coefficients of log expenditure. The reported parameters are $\theta = (\text{vec } E^x, e^x, \beta^1)$. Then $\hat{\theta} - \theta \approx G(\hat{\eta} - \eta)$ with $G = \partial g / \partial \eta'$, and $\text{Var}(\hat{\theta}) = G \text{Var}(\hat{\eta}) G'$.

Influence functions. Each component of $\hat{\eta}$ is written as a sum of contributions, $\hat{\eta} - \eta \approx \sum_c \Phi_c$, one row per cluster. The first-stage moments are sums over households, attributed to their cluster: for the coefficients of an equation, $\phi_h(\hat{\beta}) = (\tilde{X}'\Omega\tilde{X})^{-1}\tilde{x}_h\omega_h e_h$ on the households of that equation; for $\hat{\omega}_G$, $\kappa_G\omega_h m_{Gh}(e_{Gh}^1)^2/\text{df} - \hat{\omega}_G/n_G^+$; for $\hat{\chi}_G$ the same with $e^0 e^1$; for $\bar{w}_G$, $\omega_h(w_{Gh} - \bar{w}_G)/\sum \omega$. The second-stage moments have their own cluster-level contribution, $[(\tilde{y}_{Gc}^1 - m_G)(\tilde{y}_{Hc}^1 - m_H) - S_{GH}]/(n_{GH} - 1)$ over the n_{GH} clusters of the pair, and a generated-regressor term: the purged averages depend on the first-stage coefficients through $\bar{y}_{Gc}^1 = \overline{\ln v_{Gc}} - \bar{x}_{Gc}'\hat{\beta}_G^1$, so that

$$\frac{\partial S_{GH}}{\partial \beta_G^1} = -\frac{1}{n_{GH} - 1} \sum_c \tilde{x}_{Gc}(\tilde{y}_{Hc}^1 - m_H), \quad (14)$$

with $\tilde{x}_{Gc}$ the cluster mean of the regressors over the reporters of G , purged like the y 's; the analogous terms hold for $\partial S_{GH}/\partial \beta_H^1$ and for R_{GH} with respect to β_G^1 and β_H^0 . These derivatives multiply the cluster sums of the household influence functions of the coefficients. The purge and the harmonic cluster sizes are held fixed; the estimation of the purge coefficients is of second order for a residual covariance.

Aggregation. With clusters as the sampling units, $\text{Var}(\hat{\eta}) = \frac{C}{C-1} \sum_c (\Phi_c - \bar{\Phi})'(\Phi_c - \bar{\Phi})$. Under `vce(svy)` the rows are first summed within the PSUs of the design, which must contain the price clusters, and then

$$\text{Var}(\hat{\eta}) = \sum_s (1 - f_s) \frac{n_s}{n_s - 1} \sum_{p \in s} (\Phi_p - \bar{\Phi}_s)'(\Phi_p - \bar{\Phi}_s), \quad (15)$$

over strata s with n_s PSUs and finite-population correction f_s , as `svy` does for a total. The weights of the households, if the design has them, are the weights of the estimation.

Jacobian. G is computed analytically, by forward differentiation of the chain (13), (9), (7), (6), (8): a direction $d\eta$ is propagated through $dA = dS - D(d\omega)\bar{N}_+^{-1}$, $d\tilde{B} = A^{-1}(dR_f - dA\tilde{B})$, the differential of the projection (9) including that of $(I \otimes A)^{-1}$, $d\zeta$, $d\Psi = -\Psi d(\Psi^{-1})\Psi$, $d\Theta$, dE , and the same for the completed system. Applying it to the $2M^2 + 5M$ unit directions gives G exactly, at the cost of a few microseconds per direction; the test suite checks it against central differences to 10^{-5} in relative terms.

4.2 Bootstrap

`vce(bootstrap)` draws clusters — or the PSUs of `svyset`, within strata — with replacement, lets the households travel with their cluster and their weights, treats a cluster drawn twice as two clusters, and re-runs both stages on every replication; the variance is that of the replications. Deaton's programs bootstrap the second stage only, on the cluster-level file, holding the first stage fixed to save computer time (Deaton, 1997, p. 313); `shortcut` reproduces this. The shortcut assigns no variance to the expenditure and quality elasticities and omits the first-stage contribution to the price elasticities; Table 8 measures it.

5 Diagnostics

The method lives on the between-cluster variation of the unit values after the measurement-error correction. `duvmdiag`, with the syntax of `duvm`, runs the two stages without the variance and reports for each good the households and clusters that report a unit value, the harmonic cluster sizes $\bar{n}^+$ and $\bar{n}$ that enter (13), the between-cluster standard deviation of the purged unit value, the share of its variance that the correction $\hat{\omega}/\bar{n}^+$ removes, the corrected variance, the quality elasticity β^1 , the expenditure elasticity and the quality share β^1/e ; then the condition number and smallest eigenvalue of the corrected moment matrix and the relative move that symmetry imposes on B . It warns when few households report, when $\bar{n}^+ < 2$, when the correction removes more than half of the variance or makes it non-positive, when $\beta^1 < 0$ (the quality correction then raises the price elasticities instead of lowering them), when e is near zero (ζ is then unstable), when the moment matrix is ill-conditioned, and when symmetry moves B by more than half. The same report is available after estimation with `estat diagnostics`.

6 The command

6.1 Syntax

```
duvm goods [if] [in] [weight], hysize(varname) expend(varname) cluster(varname)
[region(varname) subround(varname) indcat(varlist) indcon(varlist)
csb(1) qother(#) nosymmetry vce(vcetype) hgroup(varname) dec(#) compat]
```

For each item g of *goods* the data hold `wg`, the budget share, and `luvg`, the log unit value, missing for non-buyers. `vce()` is `cluster` (linearized, the default), `svy` (linearized, design from `svyset`), `bootstrap` with sub-options `reps()`, `seed()`, `strata()`, `psu()`, `svy`, `shortcut`, or `none`. `hgroup()` re-estimates the model within each group and reports the own-price elasticities by group. The WELCOM syntax — `hweight()`, `boot()` — is accepted. After estimation, `estat diagnostics`, `estat elasticities` (with `unrestricted`, `noquality`, `uncompleted`) and `estat quality` redisplay the results; `e(b)` holds E^x row by row, e^x and β^1 , `e(V)` their variance, and every intermediate matrix is stored. A dialog box is available with `db duvm`.

6.2 Results on the Mexican cereal groups

The example data are the four cereal groups of the ENIGH 2014 — corn, wheat products, rice, other cereals — for 15,866 households in 2,622 primary sampling units, prepared as in the WELCOM manual: unit values as Laspeyres indices of the items within a group, the unit value of a non-buying household replaced by its cluster average. The model has log expenditure (current income), log household size, the age of the head, and dummies for sex and education, with the rural/urban region purged from the cluster averages; the sampling weight is used as an analytic weight.

Table 1: The data behind each good: mean budget share, households and clusters reporting a unit value, harmonic cluster sizes of the reporters ($\bar{n}^+$) and of all households ($\bar{n}$)

Good	Share (%)	Reporters	%	Clusters	$\bar{n}^+$	$\bar{n}$
Corn	2.32	15,375	96.9	2,613	3.99	4.15
Wheat	1.50	15,340	96.7	2,605	3.99	4.15
Rice	0.15	13,172	83.0	2,014	4.31	4.15
Other cereals	0.10	12,166	76.7	1,827	4.28	4.15

Table 1 shows what the method has to work with. Corn and wheat are bought by 97% of households, rice and other cereals by 77–83%; the clusters hold four households on average

in harmonic mean, so the measurement-error correction matters. Table 2 gives the first-stage parameters. The quality elasticities are small, 0.015 for corn and below 0.011 elsewhere: richer households buy slightly better cereals, not more expensive ones. The expenditure elasticities of quantity run from 0.25 for corn to 1.15 for other cereals. The quality share β^1/e is 6% for corn and about 1% for the others, and the diagonal of Ψ shows that the unit values respond to prices almost one for one.

Table 2: First stage: share and quality coefficients of log expenditure, expenditure elasticity of quantity, quality parameter ζ , quality share β^1/e and own-price response of the unit value; linearized standard errors clustered by PSU

Good	$\hat{\beta}^0$	$\hat{\beta}^1$	(s.e.)	$\hat{e}$	(s.e.)	$\hat{\zeta}$	$\hat{\zeta}\bar{w} = \hat{\beta}^1/\hat{e}$	$\hat{\psi}_{GG}$
Corn	-0.0171	0.0147	(0.0039)	0.247	(0.019)	2.581	0.060	0.951
Wheat	-0.0080	0.0042	(0.0028)	0.461	(0.019)	0.609	0.009	0.993
Rice	-0.0007	0.0001	(0.0039)	0.559	(0.036)	0.162	0.000	1.000
Other cereals	0.0002	0.0105	(0.0030)	1.150	(0.052)	9.079	0.009	0.994

Table 3: Price elasticities of quantity: quality corrected, symmetry restricted, completed system; rows are quantities, columns prices; linearized standard errors clustered by PSU in parentheses

Quantity of	p_{corn}	p_{wheat}	p_{rice}	$p_{\text{other cereals}}$	$p_{\text{all other}}$
Corn	-0.814 (0.023)	0.053 (0.013)	-0.013 (0.010)	-0.001 (0.013)	0.528 (0.031)
Wheat	0.077 (0.020)	-0.813 (0.026)	0.066 (0.015)	0.014 (0.014)	0.196 (0.036)
Rice	-0.207 (0.149)	0.642 (0.145)	-1.128 (0.070)	-0.104 (0.041)	0.238 (0.188)
Other cereals	-0.033 (0.296)	0.194 (0.215)	-0.160 (0.062)	-0.703 (0.072)	-0.448 (0.339)
All other goods	-0.003 (0.001)	-0.004 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.769 (0.001)

Table 3 is the final matrix. Own-price elasticities are -0.81 for corn, -0.81 for wheat, -1.13 for rice and -0.70 for other cereals, with standard errors of 0.02 to 0.07; the composite of all other goods, whose share is 96% of the budget, has an own-price elasticity of -0.77 that is essentially its expenditure elasticity. The cross-price elasticities of rice and other cereals, whose shares are 0.15% and 0.10% of the budget, carry standard errors of 0.1 to 0.3: small shares divide small coefficients. Table 4 follows the own-price elasticities through the treatments: the quality correction lowers the corn elasticity from -0.85 to -0.81 , in proportion to its 6% quality share, and leaves the others almost unchanged; symmetry moves rice from -1.09 to -1.13 and completion changes nothing, as it should.

Table 4: Own-price elasticities of quantity under the four treatments

Good	No quality correction	Quality, unrestricted	Quality, symmetric	Completed system
Corn	-0.855	-0.813	-0.814	-0.814
Wheat	-0.825	-0.818	-0.813	-0.813
Rice	-1.086	-1.085	-1.128	-1.128
Other cereals	-0.713	-0.709	-0.703	-0.703

Table 5 re-estimates the model within each quintile of income, as `hgroup()` does. The own-price elasticities of corn and wheat are determined to within 0.06–0.13 in four quintiles out of five, but the fourth quintile returns a rice elasticity of -2.0 with a standard error of 21: the

Table 5: Own-price elasticities by quintile of per capita current income, the model re-estimated within each quintile; linearized standard errors

Group	Corn	(s.e.)	Wheat	(s.e.)	Rice	(s.e.)	Other	(s.e.)
Quintile_1	-0.816	(0.056)	-1.039	(0.062)	-1.797	(0.450)	-1.044	(0.313)
Quintile_2	-0.893	(0.127)	-0.914	(0.065)	-1.244	(0.342)	-0.745	(0.257)
Quintile_3	-0.827	(0.221)	-0.825	(0.090)	-1.101	(0.238)	-0.738	(0.175)
Quintile_4	-1.482	(0.888)	-0.987	(1.695)	-2.000	(20.653)	-0.706	(0.723)
Quintile_5	-0.826	(0.069)	-0.554	(0.065)	-1.056	(0.240)	-0.467	(0.105)

corrected moment matrix of that group is close to singular, and the standard error says so. This is the price of cutting 2,600 clusters into five pieces of fewer than two reporting households each; `duvmdiag` warns, on every group, that the harmonic cluster size of the reporters falls below two. By decile, as the WELCOM manual reports, the same exercise gives a corn elasticity of -3.6 with a standard error of 13 in the fifth decile. A group analysis of the unit-value model needs groups that keep the clusters whole — regions, not income classes.

6.3 Timing

Table 6: Execution times in seconds (StataNow 19.5, one core)

Version	Estimation	Ten deciles	Bootstrap replication, both stages	Replication, shortcut
WELCOM <code>duvm</code> (2017)	2.36	6.10	—	0.06
<code>duvm</code> 1.0	0.16	0.42	0.14	0.02

Table 6 compares the WELCOM version, whose loop over goods calls `areg` and `egen` and whose bootstrap re-reads files from disk, with the Mata engine. One estimation takes a sixth of a second instead of 2.4 s; a bootstrap replication of both stages takes 0.15 s, so that 1,000 replications take about two and a half minutes; Deaton’s shortcut, which repeats only the second stage, takes two hundredths of a second, three times less than the WELCOM bootstrap that implements the same shortcut.

7 Validation of the variance

The linearized variance is validated against the bootstrap of both stages, 400 replications, in the order recommended for survey estimators: first with the clusters as the sampling units and no weights, then with the sampling weights, then with a stratified design declared to `svyset` (PSUs, strata = the quintiles of income, `pweight`). Table 7 reports the own-price standard errors under the two methods and, over the 34 parameters of `e(b)`, the minimum, median and maximum of the ratio of the linearized to the bootstrap standard error. The median ratio is within two percent of one in the three designs, and every ratio lies between 0.89 and 1.10, the spread expected from 400 replications.

Table 8 measures what Deaton’s shortcut leaves out. Bootstrapping the second stage alone understates the standard error of the corn elasticity by 17% (ratio 1.21), that of the composite good by 10%, and those of the other cereals by 2–3%: the first-stage contribution is small for goods whose quality share is small, and largest for corn, whose quality share is 6%. The shortcut gives no standard error at all to the expenditure and quality elasticities.

Table 7: Linearized against bootstrap standard errors (400 replications of both stages): own-price elasticities, and the distribution of the ratio over the 34 parameters

Design	Standard error	Corn	Wheat	Rice	Other	All other	Ratio over the 34 parameters		
							min	median	max
Simple random sampling of clusters	Linearized	0.0212	0.0256	0.0690	0.0663	0.0011	0.962	1.019	1.099
	Bootstrap	0.0206	0.0262	0.0697	0.0677	0.0010			
Sampling weights (aweight)	Linearized	0.0230	0.0259	0.0696	0.0718	0.0011	0.894	0.987	1.066
	Bootstrap	0.0234	0.0272	0.0682	0.0733	0.0011			
Survey design (svyset, strata, pweight)	Linearized	0.0230	0.0259	0.0696	0.0718	0.0011	0.944	0.993	1.069
	Bootstrap	0.0228	0.0253	0.0708	0.0682	0.0011			

Table 8: Bootstrap of both stages against the shortcut that resamples the cluster-level file only, 400 replications

Bootstrap	Own-price elasticity					Expenditure elasticity			
	Corn	Wheat	Rice	Other	All other	Corn	Wheat	Rice	Other
Both stages	0.0241	0.0260	0.0704	0.0702	0.0011	0.0181	0.0200	0.0365	0.0511
Second stage only (shortcut)	0.0199	0.0253	0.0685	0.0690	0.0010	0.0000	0.0000	0.0000	0.0000
Ratio	1.209	1.027	1.028	1.016	1.115	–	–	–	–

8 Conclusion

`duvm` implements Deaton’s unit-value model as the book states it, with the inference the method had been lacking: a variance that accounts for both stages and for the survey design, a bootstrap of the whole procedure to check it, and a diagnostic of the price signal. The estimator is fast enough for resampling and for estimation by group. The formulas of the earlier implementation remain available under `compat`, so that results computed with it can be reproduced and the effect of each correction can be seen; the appendix documents them.

A The published code and the book

The Stata programs published with [Deaton \(1997\)](#) implement chapter 5 with four departures from its equations, which the WELCOM `duvm` inherited.

1. **The quality parameter.** Program `mormat` computes $\zeta_G = \beta_G^1 / (\beta_G^0 + 1 - \beta_G^1 \bar{w}_G)$, with a 1 where eq. (5.92) has $\bar{w}_G$. Since $\bar{w}_G$ is a few percent, this makes $\zeta_G \approx \beta_G^1$ and reduces the quality correction to almost nothing; with the book’s formula $\zeta_{\text{corn}} = 2.6$ against 0.015. The identity (5), $\Psi = I + D(\beta^1)D(e)^{-1}E$, holds to 10^{-16} with the book’s ζ and fails by 0.05 with the code’s.
2. **The completed system.** Program `complet` builds the extended ζ^x as $[\bar{w}(1 - \beta^1) + \beta^0] / \beta^1$, the reciprocal of (7), through `syminv(diag(b1))`; `syminv` also zeroes any row whose pivot is not positive, so a good with a negative quality elasticity loses its correction in the completed system. This happens in subsamples: in the first decile of the Mexican data three of the four β^1 are negative.
3. **The last row of Θ^x .** `complet` appends the column sums of Θ^x where eq. (5.94) requires their negatives, so that the columns of the completed matrix do not add to zero.
4. **Cluster sizes with weights (WELCOM only).** The size of a cluster is taken as the sum of its sampling weights, several thousand instead of about four, so that the correction $\hat{\omega} / \bar{n}^+$ in (13) vanishes.

Two further details are matters of convention rather than error: WELCOM obtains $\hat{\chi}$ from an auxiliary within-cluster regression of e^1 on e^0 and the regressors rather than from (5.82), and its

weighted cluster mean of the unit values divides the average over the reporters by the average weight of all households; the value of the region for a cluster is that of an arbitrary household rather than the first non-missing one.

Table 9 moves from the WELCOM formulas to the book one correction at a time, on the weighted model of Section 6. The quality parameter of eq. (5.92) alone changes little, because the completed system still inverts it; once the completion uses ζ^x itself the quality correction becomes effective and the corn elasticity moves from -0.88 to -0.84 . The effective cluster sizes restore the measurement-error correction and move it to -0.82 ; the sign of the last row of Θ^x touches only the composite good, whose own-price elasticity goes from -0.25 to -0.77 ; the remaining conventions matter at the third decimal.

Table 9: From the WELCOM formulas to the book, one correction at a time: own-price elasticities of the completed, symmetry-restricted system

Formulas	Corn	Wheat	Rice	Other	All other
WELCOM formulas (all)	-0.8777	-0.7996	-1.1589	-0.7095	-0.2546
+ ζ of eq. (5.92)	-0.8834	-0.8006	-1.1581	-0.7114	-0.2546
+ completion with ζ^x , not $1/\zeta^x$	-0.8410	-0.8129	-1.1688	-0.7050	-0.7432
+ last row of $\Theta^x = -\text{colsum}$	-0.8410	-0.8129	-1.1688	-0.7050	-0.7698
+ effective cluster sizes	-0.8185	-0.8124	-1.1525	-0.6951	-0.7694
+ χ from eq. (5.82)	-0.8188	-0.8124	-1.1474	-0.6930	-0.7694
+ cluster means over the observed households	-0.8188	-0.8124	-1.1474	-0.6930	-0.7694
+ region = first non-missing household (the book)	-0.8138	-0.8129	-1.1281	-0.7033	-0.7690

Option `compat` reproduces the WELCOM version, and `compatflags()` selects individual departures. Table 10 compares `duvm`, `compat` with the frozen output of the WELCOM program on five configurations of the Mexican data: the largest relative difference over every intermediate matrix and over the elasticities is of the order of 10^{-7} , the precision of the float variables the WELCOM pipeline creates.

Table 10: `duvm`, `compat` against the WELCOM program: largest relative difference

Case	Moments and coefficients	Elasticities
weights, region	3.0e-09	4.8e-08
unweighted, region	1.0e-08	8.7e-08
weights, no region	3.6e-09	4.4e-08
weights, region, <code>csb(1)</code>	3.0e-09	7.8e-07
weights, region, continuous controls	4.3e-09	6.5e-08

B Tests

The package ships with the scripts of the `replication/` folder, one per table of this note, run by `master.do`. The development test suite (not distributed) freezes the WELCOM output on six configurations and checks `compat` against it; checks the identities (5.87), (5.93), (5.94) and (5.98) on the default estimates to 10^{-12} ; checks the analytic Jacobian against central differences; checks the linearized variance against the bootstrap under the three designs of Section 7; and exercises the bootstrap options, `hgroup()`, `estat` and `duvmdiag`.

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